

Digital Image Processing

Lecture # 3

Connected Component Analysis & Spatial Enhancement-I

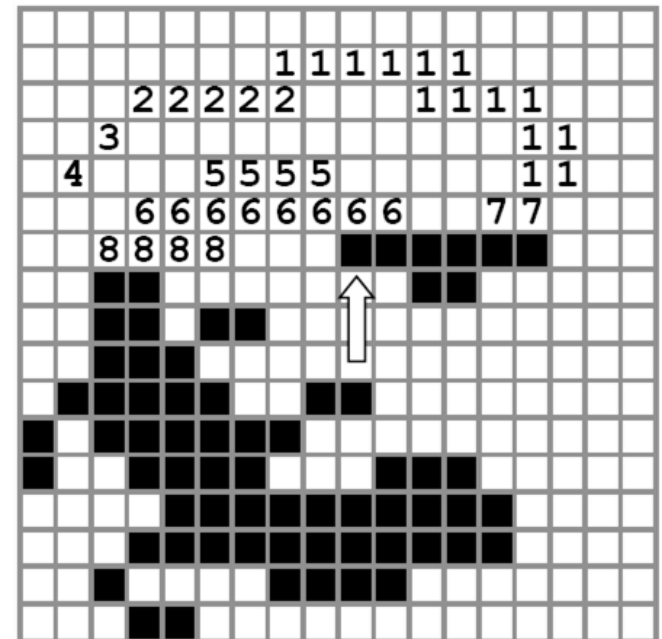
Connected Component Analysis

CC labeling – 4 Connectivity

- ◆ Process the image from left to right, top to bottom:
 - 1.) If the next pixel to process is 1
 - i.) If only one of its neighbors (top or left) is 1, copy its label.
 - ii.) If both are 1 and have the same label, copy it.
 - iii.) If they have different labels
 - Copy the label from the left.
 - Update the equivalence table.
 - iv.) Otherwise, assign a new label.



Pass 1

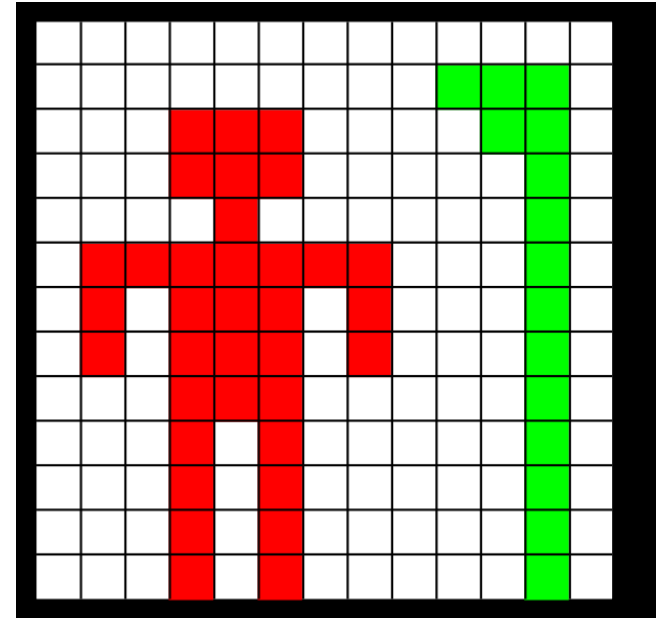
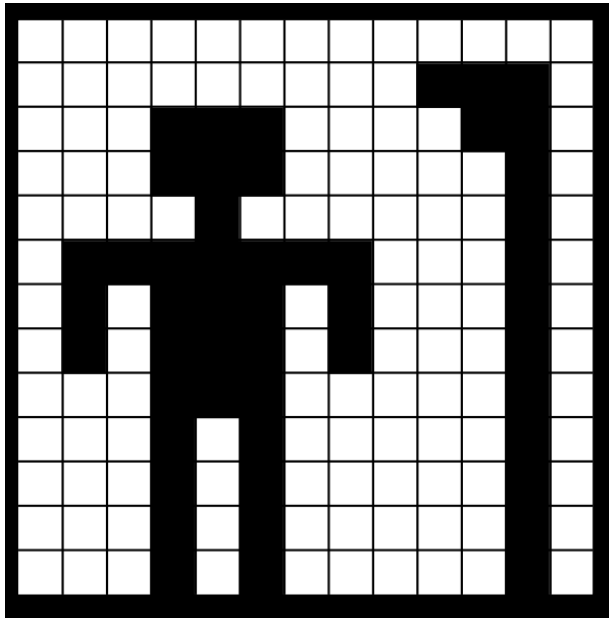


- ◆ Re-label with the smallest of equivalent labels

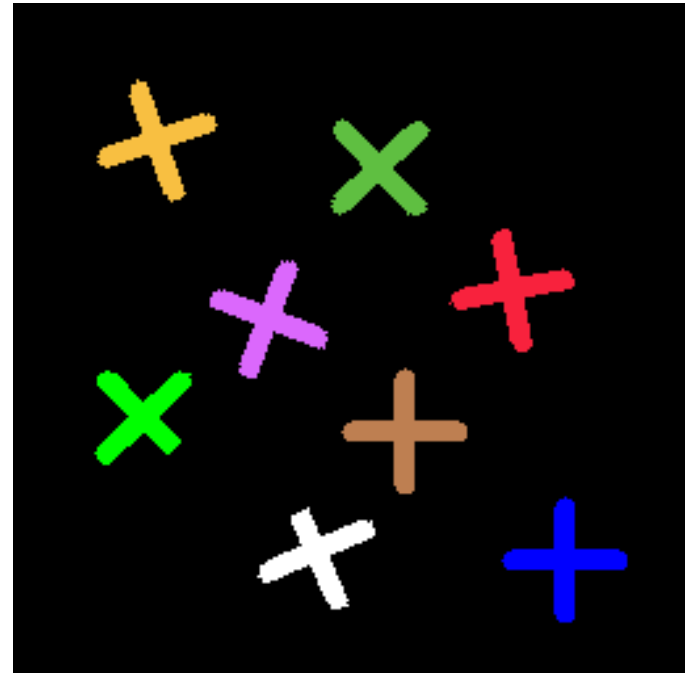
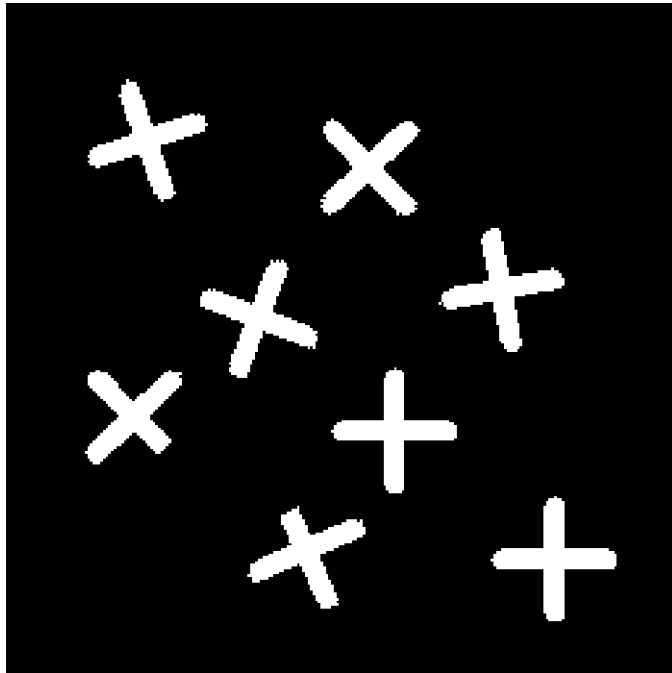
Pass 2

$\{1, 3, 4, 5\}$	$2, 7$
$\{6, 8\}$	

CC labeling – 4 Connectivity



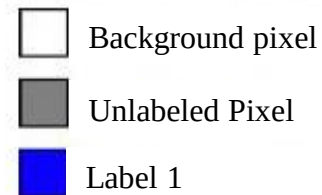
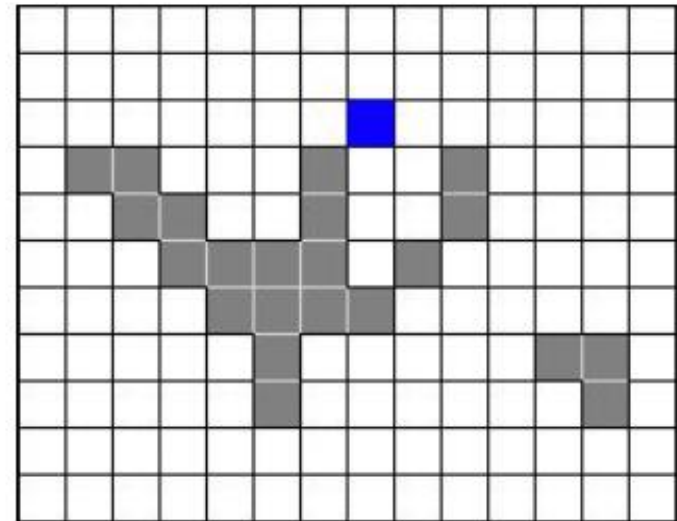
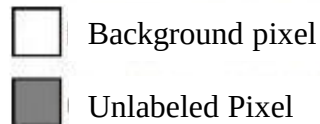
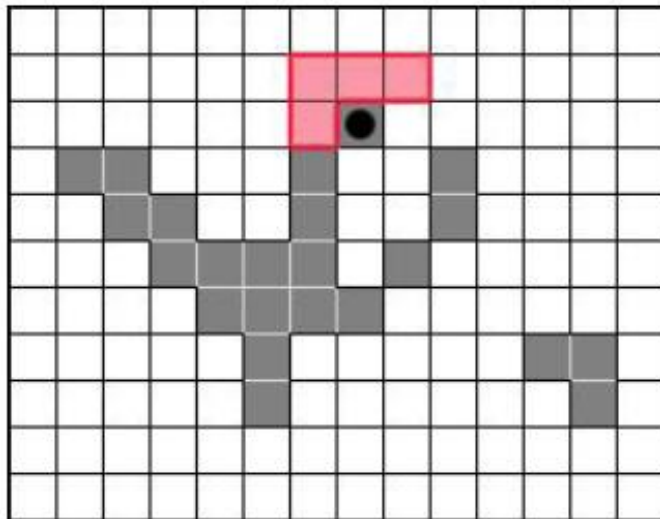
CC labeling – 4 Connectivity



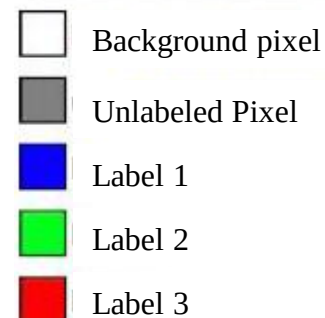
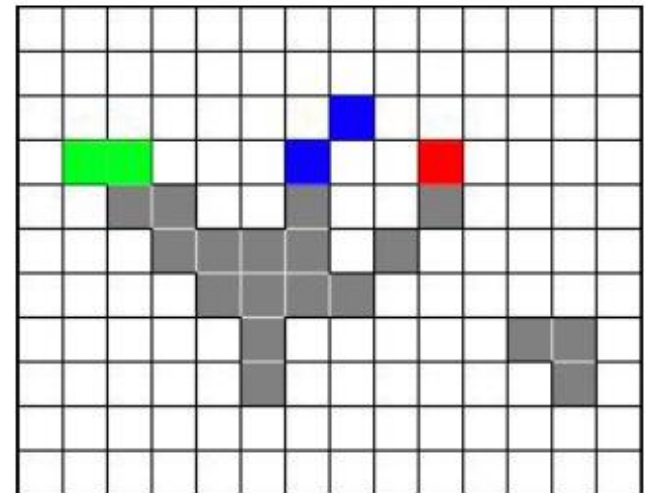
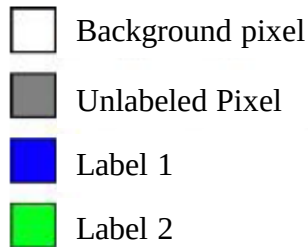
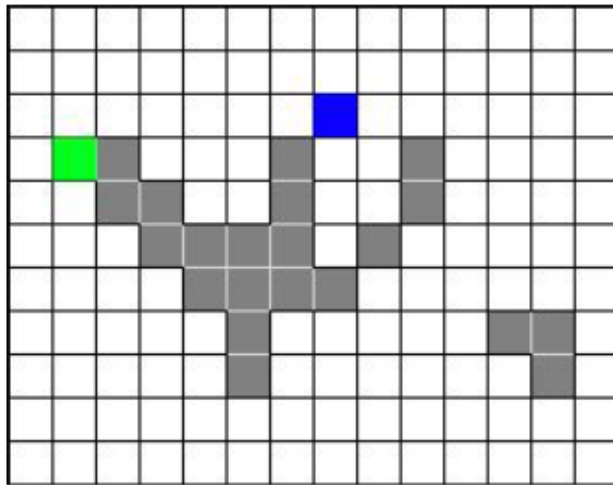
CC labeling – 8 Connectivity

Same algorithm but examine also the upper diagonal neighbors of p

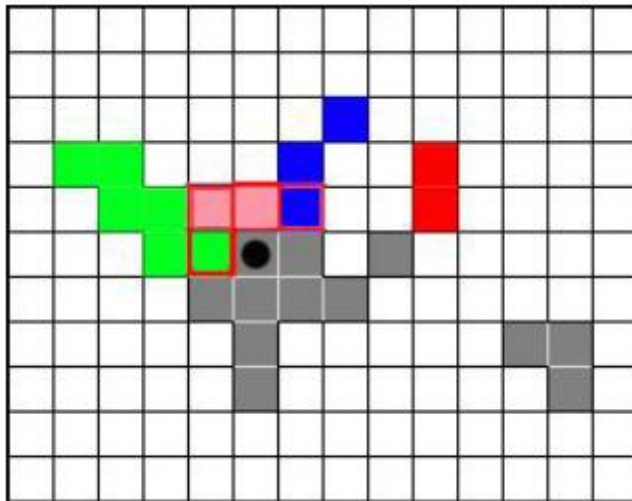
CC labeling – 8 Connectivity



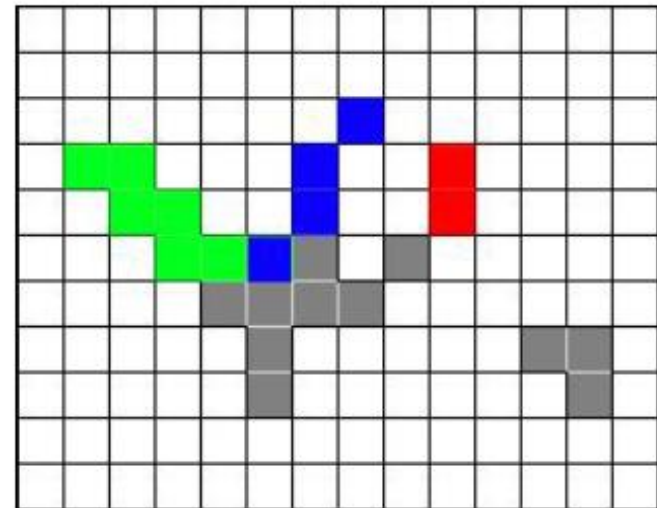
CC labeling – 8 Connectivity



CC labeling – 8 Connectivity



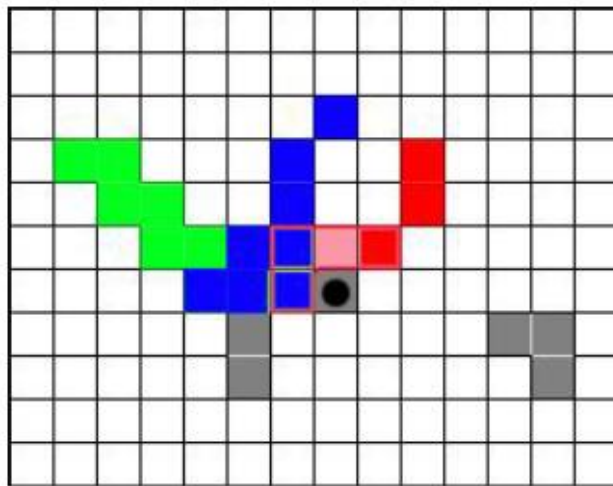
-  Background pixel
-  Unlabeled Pixel
-  Label 1
-  Label 2
-  Label 3



-  Background pixel
-  Unlabeled Pixel
-  Label 1
-  Label 2
-  Label 3

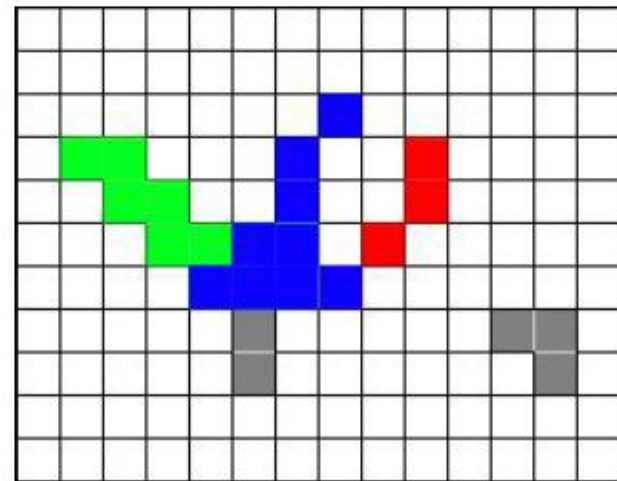
EQUIVALENCE TABLE	
	

CC labeling – 8 Connectivity



- Background pixel
- Unlabeled pixel
- Label 1
- Label 2
- Label 3

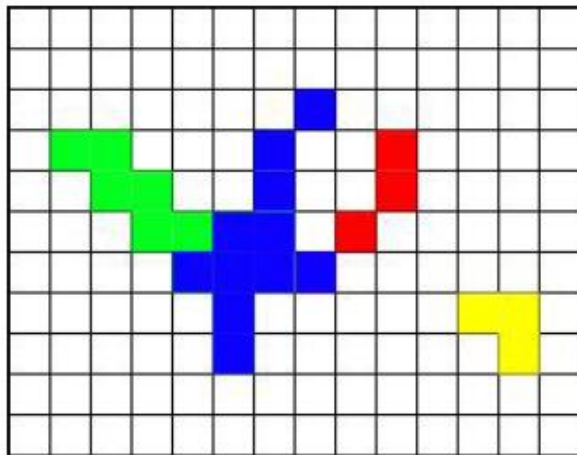
EQUIVALENCE TABLE	
Label 1	Label 2



- Background pixel
- Unlabeled pixel
- Label 1
- Label 2
- Label 3

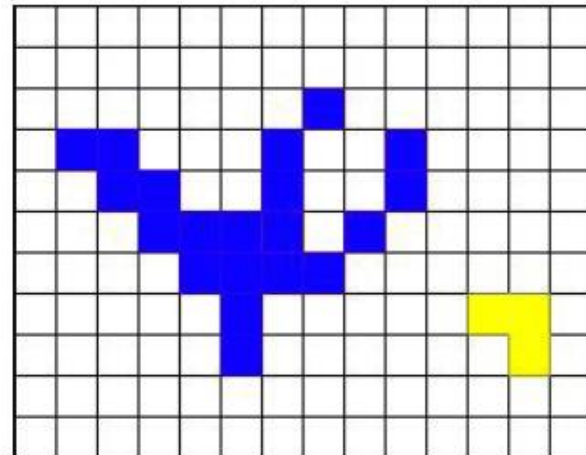
EQUIVALENCE TABLE		
Label 1	Label 2	Label 3

CC labeling – 8 Connectivity



-  Background pixel
-  Unlabeled pixel
-  Label 1
-  Label 2
-  Label 3
-  Label 4

EQUIVALENCE TABLE		
		



-  Background pixel
-  Unlabeled pixel
-  Label 1
-  Label 2
-  Label 3
-  Label 4

EQUIVALENCE TABLE		
		

Image Enhancement

Image Enhancement



Image Enhancement

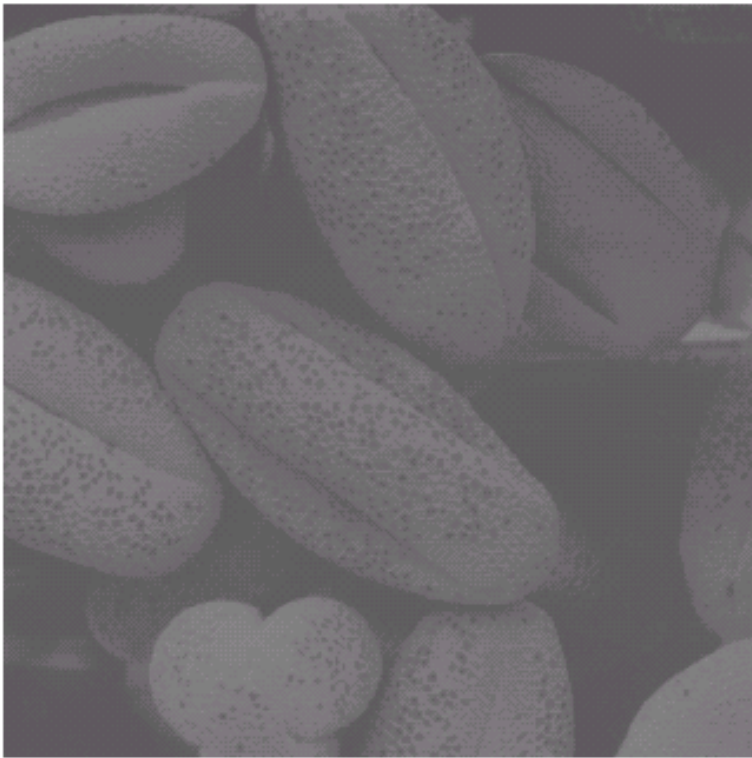


Image Enhancement

Process an image so that the result is more suitable than the original image for a **specific application**

- ◆ Image Enhancement Methods
 - **Spatial Domain**: Direct manipulation of pixels in an image
 - **Frequency Domain**: Process the image by modifying the Fourier transform of an image

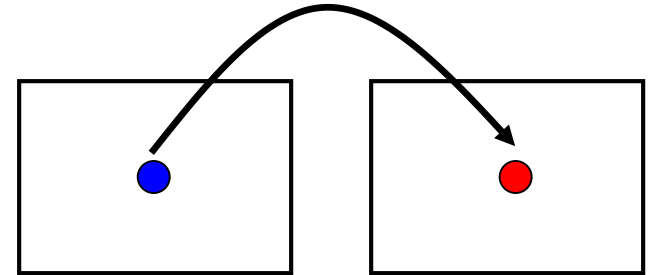
This Chapter – Spatial Domain



Types of image enhancement operations

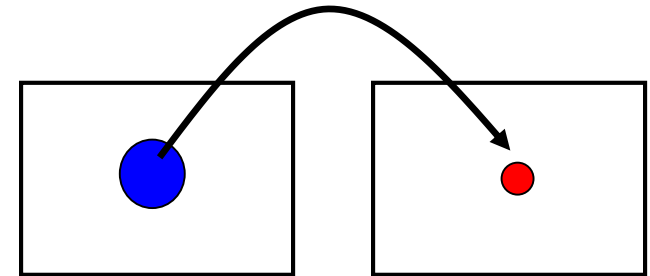
- ◆ Point/Pixel operations

Output value at specific coordinates (x,y) is dependent only on the input value at (x,y)



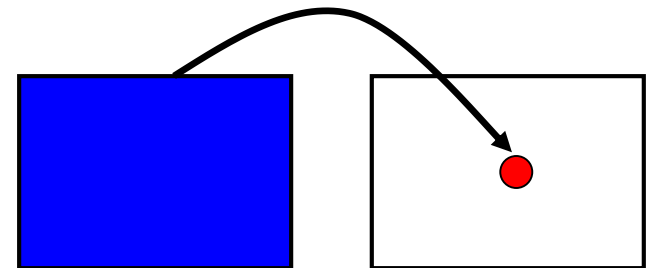
- ◆ Local operations

The output value at (x,y) is dependent on the input values in the neighborhood of (x,y)



- ◆ Global operations

The output value at (x,y) is dependent on all the values in the input image



Basic Concepts

- ◆ Most spatial domain enhancement operations can be generalized as:

$$g(x, y) = T[f(x, y)]$$

$f(x, y)$ = the input image

$g(x, y)$ = the processed/output image

T = some operator defined over some neighbourhood of (x, y)

Point Processing

- ◆ In a digital image, point = pixel
- ◆ Point processing transforms a pixel's value as function of its value alone;
- ◆ It does not depend on the values of the pixel's neighbors.

Point Processing

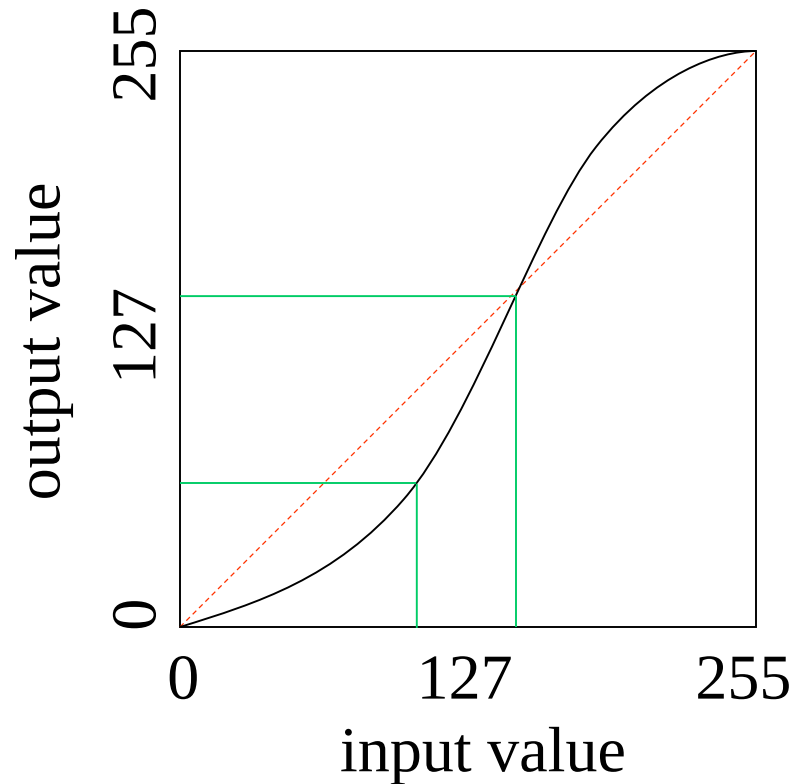
- ◆ Neighborhood of size 1x1:
- ◆ g depends only on f at (x,y)
- ◆ T : Gray-level/intensity transformation/ mapping function

$$s = T(r)$$

- r = gray level of f at (x,y)
- s = gray level of g at (x,y)

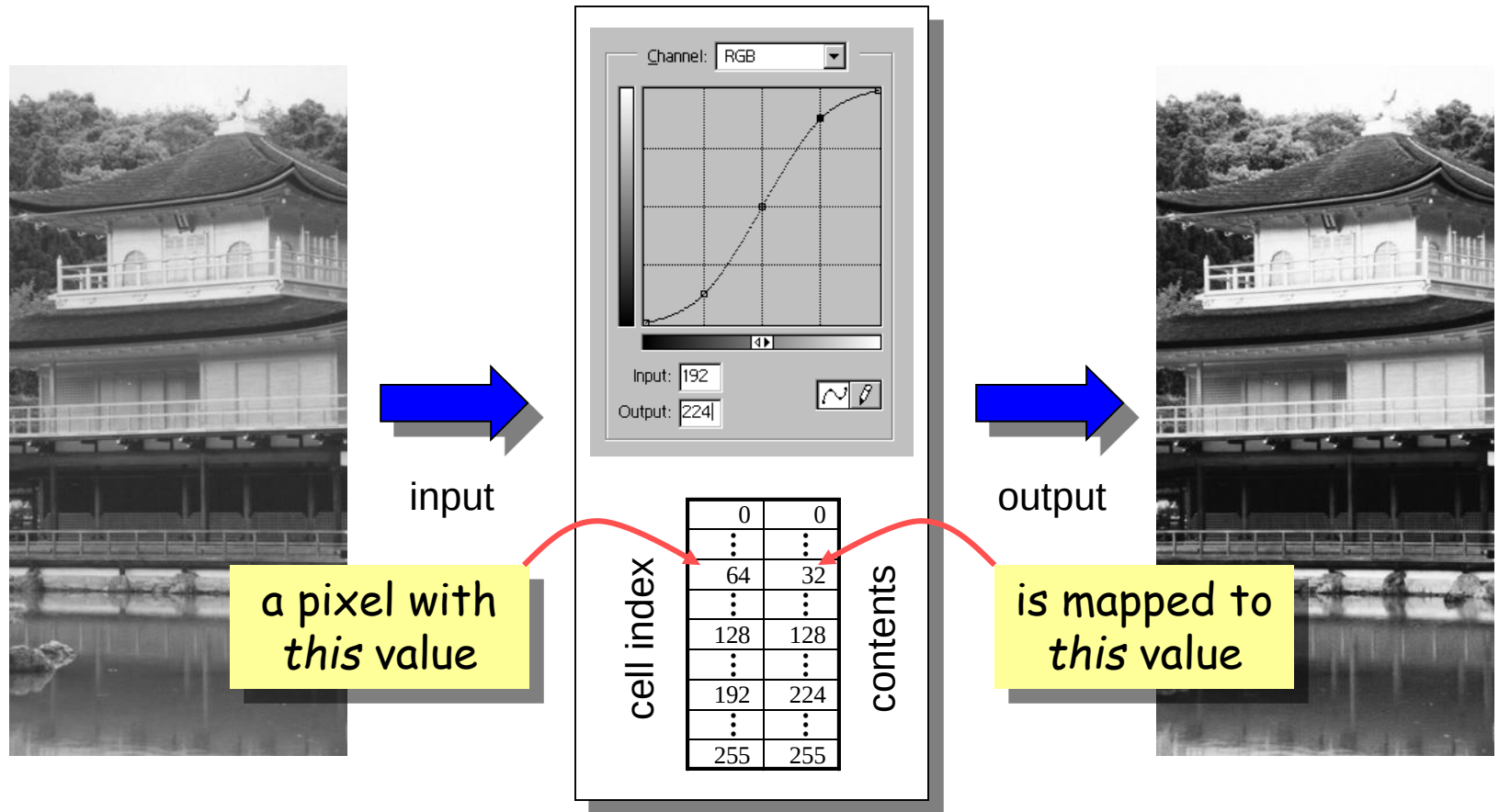
Point Processing using Look-up Tables

A look-up table (LUT) implements a functional mapping.

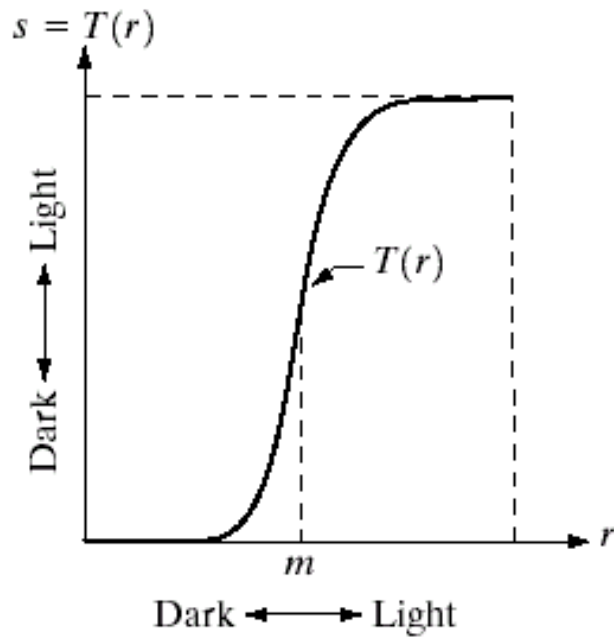


<i>E.g.:</i>	index	value
	...	...
	101	64
	102	68
	103	69
	104	70
	105	70
	106	71
	...	...
	input	output

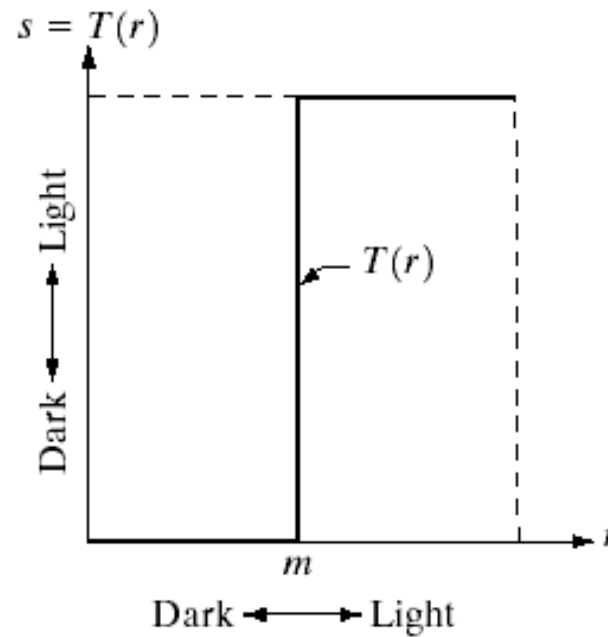
Point Processing using Look-up Tables



POINT PROCESSING



Contrast Stretching



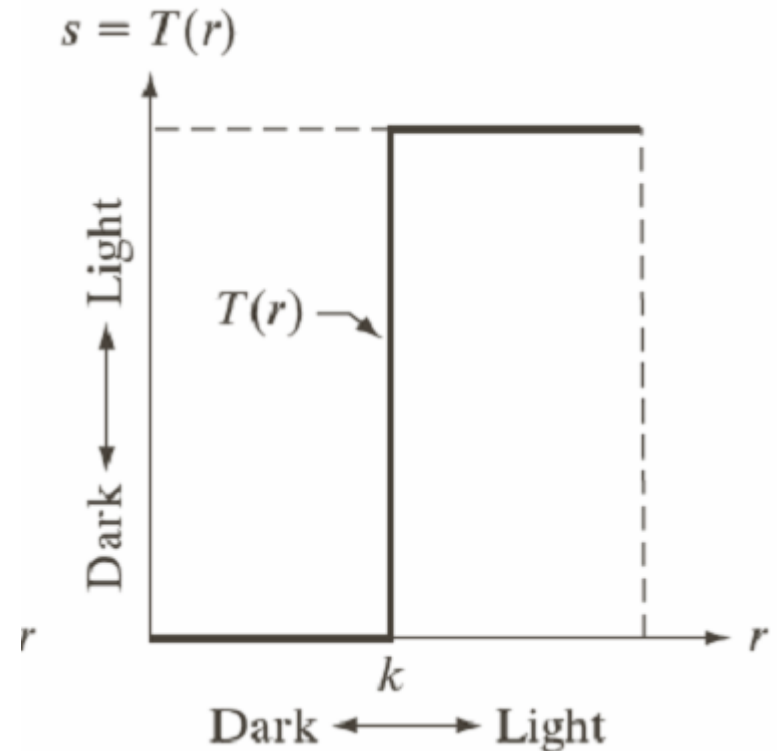
Thresholding

a b

FIGURE 3.2 Gray-level transformation functions for contrast enhancement.

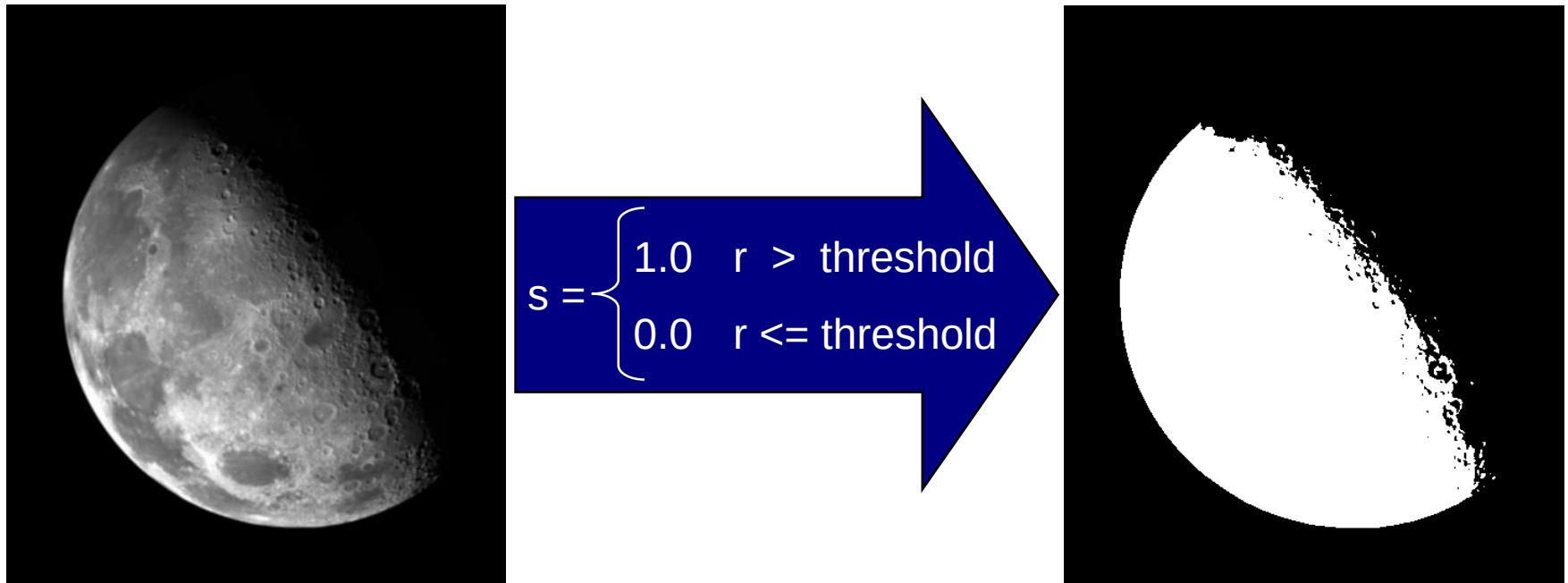
Point Processing Example: Thresholding

$$s = \begin{cases} 1.0 & r > \text{threshold} \\ 0.0 & r \leq \text{threshold} \end{cases}$$



Point Processing Example: Thresholding

- ◆ Segmentation of an object of interest from a background



Point Processing Example: Intensity Scaling

$$s = T(r) = a \cdot r$$

Original image



$f(x,y)$

Scaled image



$a \cdot f(x,y)$

Point Processing Transformations

- ◆ There are many different kinds of grey level transformations

- ◆ Three of the most common are shown here

- Linear

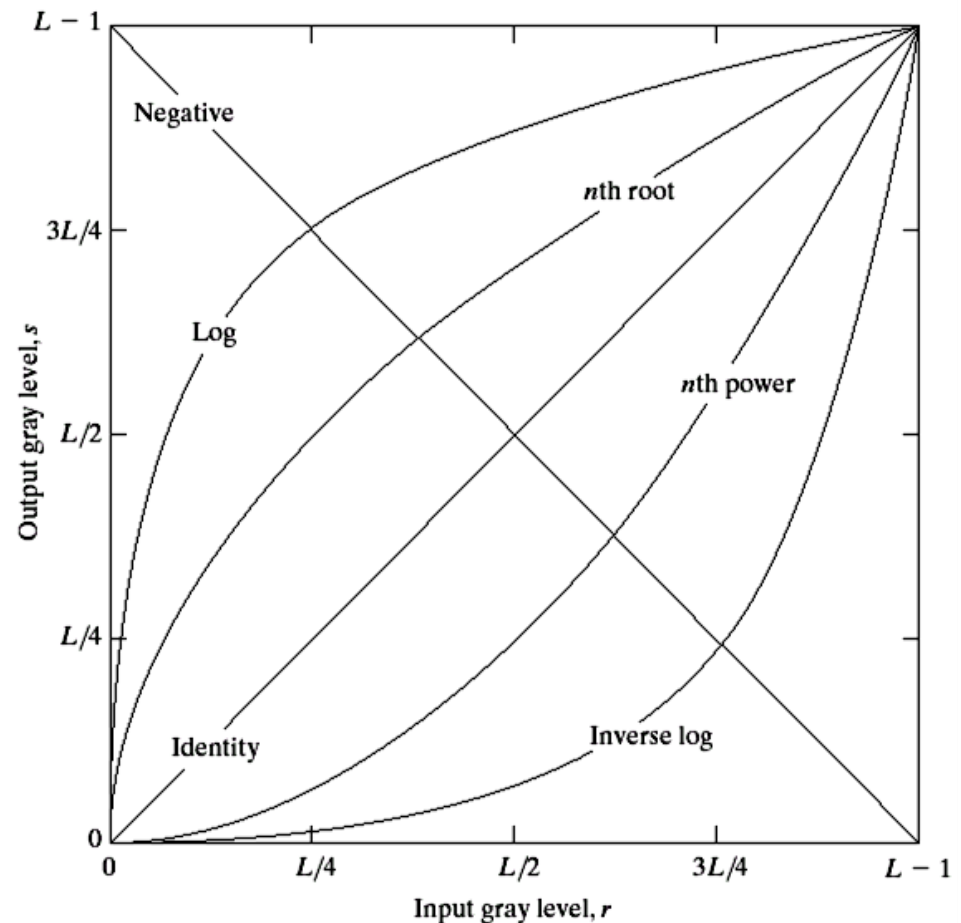
- Negative/Identity

- Logarithmic

- Log/Inverse log

- Power law

- n^{th} power/ n^{th} root



Point Processing Example: Negative Images

- ◆ Reverses the gray level order
- ◆ For L gray levels, the transformation has the form:

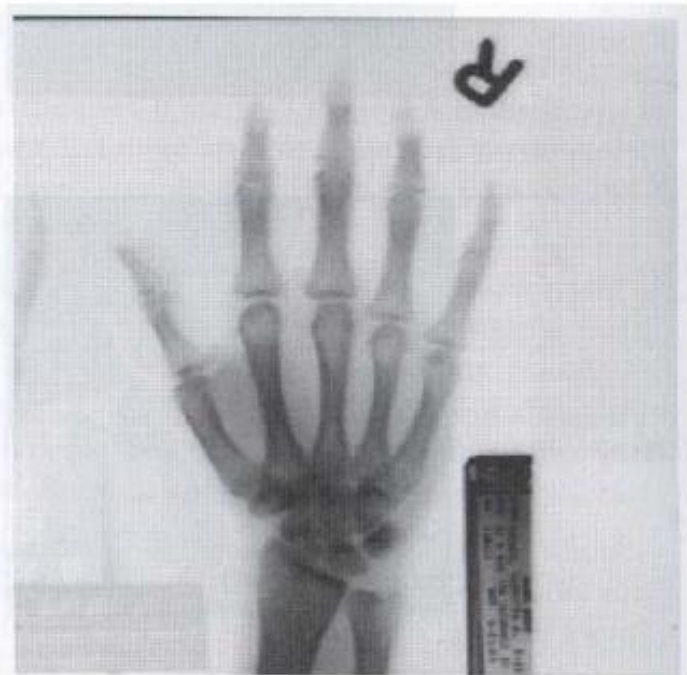
$$s = (L - 1) - r$$

- ◆ Negative images are useful for enhancing white or grey detail embedded in dark regions of an image

Point Processing Example: Negative Images



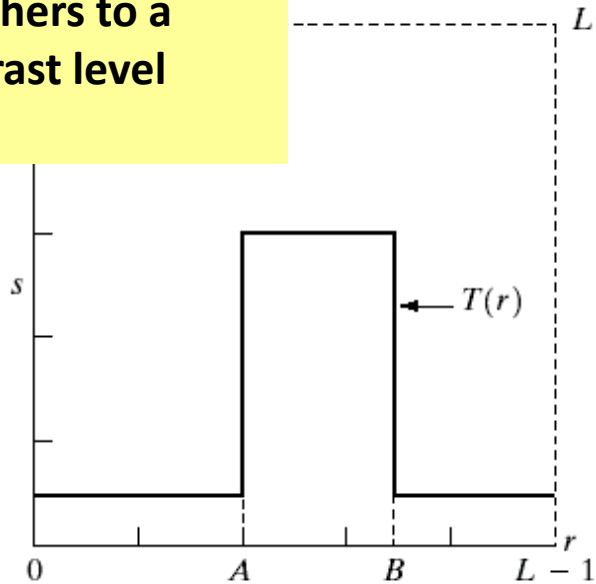
Input image (X-ray image)



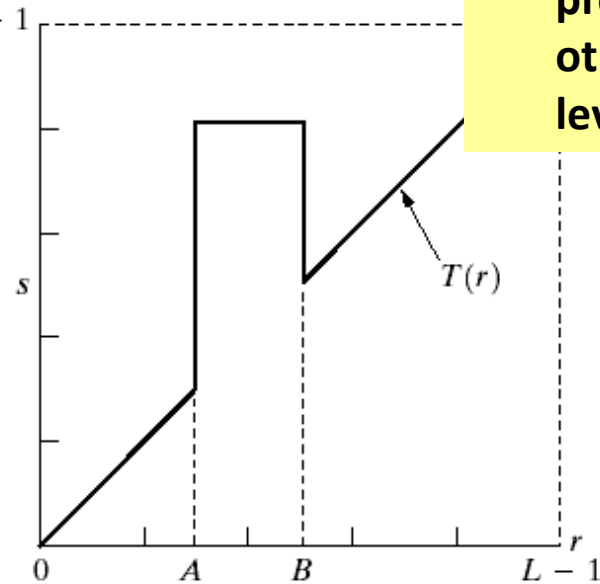
Output image (negative)

Grey Level Slicing

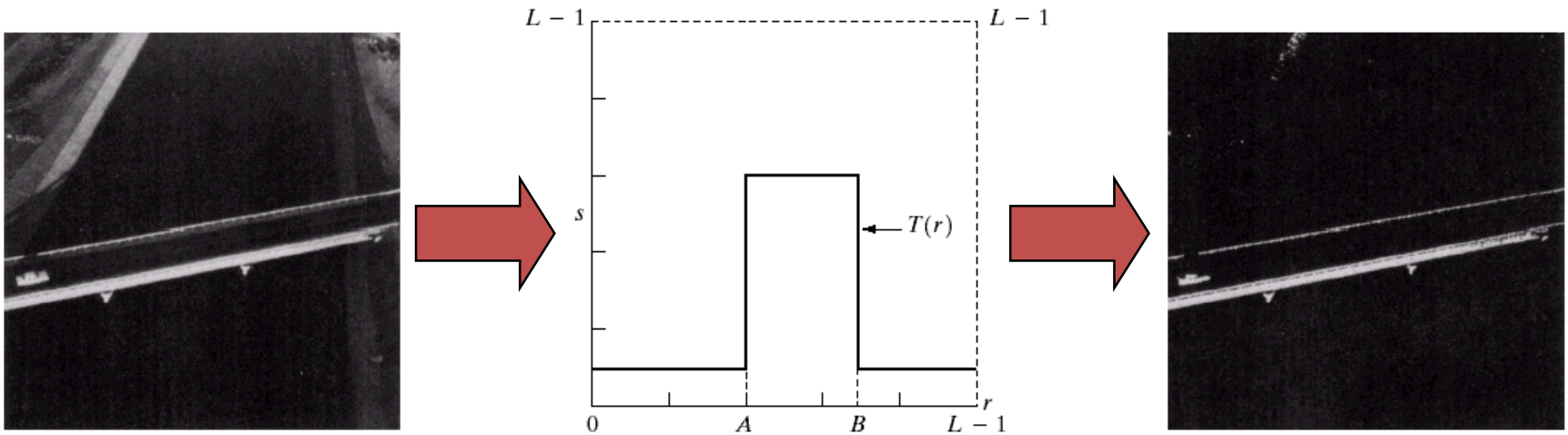
Highlights range $[A,B]$ of gray levels and reduces all others to a contrast level



Highlights range $[A,B]$ but preserves all other gray levels



Grey Level Slicing



Readings from Book (3rd Edn.)

- 2.5 Basic Relation between Pixels
- 3.1 Image Enhancement & Transformations
- 3.2 Basic Intensity Transformation Functions



Acknowledgements

- ◆ Statistical Pattern Recognition: A Review – A.K Jain et al., PAMI (22) 2000
- ◆ Pattern Recognition and Analysis Course – A.K. Jain, MSU
- ◆ *Pattern Classification*” by Duda et al., John Wiley & Sons.
- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Machine Vision: Automated Visual Inspection and Robot Vision”, David Vernon, Prentice Hall, 1991
- ◆ www.eu.aibo.com/
- ◆ Advances in Human Computer Interaction, Shane Pinder, InTech, Austria, October 2008